May Learning Hub

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Grade 11 Geography

7. Mass Movements

Learning Guide

1. What is Mass Movement?

- **Mass movement** is also called **mass wasting**.
- It is the **downward movement of weathered material under the force of gravity**.
- The movement may be:
  - **Slow** → e.g. soil creep
  - **Rapid** → e.g. landslides and mudflows
- Material may move downslope by:
  - **Creeping**
  - **Sliding**
  - **Flowing**
  - **Falling**

📌 ***Key idea:*** *Gravity is the main force responsible for moving material downslope.*

2. What Causes Mass Movement?

- Mass movement occurs when the force of **gravity becomes greater than the resistance of the slope**.
- When this happens, the slope can no longer remain stable and **slope failure** occurs.

Slope Failure

- Happens when **gravitational pull exceeds the internal friction** holding slope material together.
- The slope then collapses or begins to move downslope.

3. Factors Influencing Mass Movement

Several natural conditions determine whether a slope will remain stable or fail.

1. Slope Gradient

- A **steeper slope** allows material to move more quickly.
- Therefore, steep slopes are generally more prone to mass movement.

2. Rock Structure

- Weaknesses in rocks make movement easier.
- Examples include:
  - **Joints**
  - **Bedding planes**
- Material may slide along these weaknesses.

3. Vegetation

- Vegetation helps to **stabilise slopes**.
- Plant roots anchor soil.
- Vegetation also removes some water from the soil.
- This reduces the likelihood of slope failure.

4. Soil Type

- Some soils move more easily when wet.
- **Saturated clay soils** are especially likely to move downslope.

5. Rainfall

- Rain adds water and **weight to the slope**.
- This increases instability and may trigger movement.

6. Earthquakes

- Earthquake vibrations can:
  - Loosen rocks.
  - Dislodge material.
  - Trigger movement on unstable slopes.

🚧 4. Human Activities that Trigger Mass Movements

Human activities can weaken slopes and increase the chance of slope failure.

Building on Steep Slopes

- Buildings add extra **weight** to the slope.

Road Construction and Blasting

- Cutting and blasting into slopes can **weaken the slope structure**.

Quarrying

- Quarrying near the **base of a slope** may remove support.
- This can cause the slope above to collapse.

Deforestation and Overgrazing

- Remove vegetation.
- This removes the roots that normally **anchor the soil**.

Poor Farming Practices

- Ploughing **down the slope** loosens soil.
- Loose material is more easily moved.

Dumping Debris

- Adds loose and unstable material to slopes.
- This can increase the risk of movement.

🔄 5. Types of Mass Movement

From **slow to fast**:

1. Soil creep
2. Solifluction
3. Earthflow
4. Mudflow
5. Slumping
6. Landslide
7. Rockfall

🐌 6. Soil Creep

- **Soil creep** is the **slowest form of mass movement**.
- Movement is approximately **1 cm per year**.
- It involves very slow downslope movement of soil particles.

Cause

- Expansion and contraction of soil grains.
- This may occur under:
  - **Dry conditions**
  - **Cold conditions**

Where It Occurs

- **Crest slopes**
- **Talus slopes**

Feature Formed

- **Terracettes** develop.
- These are small **step-like features** on the slope.

Signs of Soil Creep

- Bent fences.
- Tilting trees or poles.
- Cracked walls.

📌 ***Key idea:*** *Soil creep is extremely slow, but its effects become visible over time.*

❄️ 7. Solifluction

- **Solifluction** is the **slow flow of water-saturated soil** downslope.

Cold Areas

- The upper soil layer **thaws**.
- The subsoil remains **frozen**.
- Water cannot easily move downward, so the saturated upper soil slowly flows over the frozen layer.

Arid Areas

- Solifluction may also occur after rainfall.

Characteristics

- Forms **tongue-like lobes of soil**.
- Usually occurs on **talus slopes**.

🌱 8. Earthflow

- **Earthflow** is the movement of **wet clay** down a slope.
- It is more rapid than soil creep and solifluction.

Characteristics

- Forms **tongue-shaped lobes**.
- May continue for:
  - Hours.
  - Or even years.
- Occurs mainly on:
  - **Talus**
  - **Pediment**

📌 ***Key idea:*** *Earthflow involves wet clay moving downslope in a lobe-like mass.*

🌧️ 9. Mudflow

- A **mudflow** is the **rapid movement of saturated fine particles**.
- It is particularly associated with **arid areas after heavy rainfall**.

Characteristics

- Fine material becomes saturated with water.
- It moves rapidly downslope.
- It forms **mud streams in definite channels**.

Where It Occurs

- **Talus slopes**
- **Pediments**

📌 ***Key idea:*** *Mudflow = fine material + lots of water + rapid downslope movement.*

🔄 10. Slumping

- **Slumping** is also called a **rotational landslide**.
- A block of land slides downslope along a **curved surface**.

Characteristics

- The moving block may **tilt backwards**.
- Common on **steep slopes**.
- Often leaves behind a **curved scar**.

📌 ***Key idea:*** *Slumping involves rotation along a curved surface.*

⛰️ 11. Landslides

- A **landslide** is the **sudden and rapid movement of material downslope**.
- Common where **bedding planes are parallel to the slope angle**.

Characteristics

- Material slides along a relatively flat or **planar surface**.

- Leaves behind a **planar rupture surface**.
- Common in:
    - **Jointed rocks**
    - **Layered rocks**

📌 ***Key idea:*** *Landslide = rapid sliding along a relatively straight surface.*

🪨 12. Rockfalls

- **Rockfalls** occur on **cliffs or very steep rock faces**.
- They are triggered when weathering weakens rock.

Causes

- **Physical weathering**
- **Chemical weathering**

Process

1. Rock becomes weakened.
2. Pieces detach from the cliff.
3. Rocks fall **vertically downward**.
4. Fallen rocks collect at the base.

Result

- Material accumulates on the **talus slope**.

Common Locations

- Mountains.
- Sea cliffs.

📌 ***Key idea:*** *Rockfall = weathered rock breaks from a steep face and falls vertically.*

📊 13. Comparison of the Main Types of Mass Movement

Type	Speed	Main Conditions	Slope Area
Soil creep	Very slow	Dry/cold conditions, subtle tilting	Crest, talus
Solifluction	Slow	Frozen or water-saturated soil	Talus
Earthflow	Moderate	Wet clay, forms lobes	Talus, pediment
Mudflow	Rapid	Arid area + heavy rain	Talus, pediment
Slumping	Fast	Curved sliding surface, steep slope	Steep slopes
Landslide	Fast	Planar slide, layered rock	Slope face
Rockfall	Fast	Weathered cliffs	Talus/base

🧠 14. How to Tell the Types Apart

Soil Creep
Very slow + leaning trees/fences + terracettes.
Solifluction
Saturated soil + slow flow + cold/wet conditions.
Earthflow
Wet clay + tongue-shaped lobe.
Mudflow
Fine saturated material + heavy rain + channelised flow.
Slumping
Block rotates along a **curved** surface.
Landslide
Rapid movement along a **planar/straight** surface.
Rockfall
Rocks break from cliffs and **fall vertically**.

🧠 15. Final Exam Summary

- **Mass movement / mass wasting** is the downward movement of weathered material due to **gravity**.
- It occurs when:
  Gravity > slope resistance → slope failure.
- Factors influencing mass movement include:
  - Slope gradient.
  - Rock structure.
  - Vegetation.
  - Soil type.
  - Rainfall.
  - Earthquakes.
- Human activities can increase instability through:
  - Building.
  - Road construction and blasting.
  - Quarrying.
  - Deforestation.
  - Overgrazing.
  - Poor farming.
  - Dumping debris.
- **Soil creep** is the slowest type.
- **Solifluction** is slow movement of saturated soil.
- **Earthflow** involves wet clay forming lobes.
- **Mudflow** is rapid movement of saturated fine material.
- **Slumping** occurs along a curved surface.
- **Landslides** move rapidly along a planar surface.
- **Rockfalls** involve rocks falling vertically from steep cliffs.

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